

Full Report: Refined 3D Index of 2

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1 Manifold Summary

- Manifold: 5
- Number of tetrahedra: $n = 3$
- Number of cusps: $r = 1$
- Independent easy edges: 0
- Hard edges (fugacity variables): 2
- q -order (half): 10 (q -order: 10/2)

2 SnaPy Gluing Equations

The gluing equation matrix has rows:

- Rows 0...2: edge equations
- Rows 3, 5, ...: meridian equations (per cusp)
- Rows 4, 6, ...: longitude equations (per cusp)

Columns are ordered as $Z_1, Z'_1, Z''_1, Z_2, Z'_2, Z''_2, \dots, Z_3, Z'_3, Z''_3$.

$$G_{\text{glue}} = \begin{pmatrix} 1 & 0 & 1 & 1 & 2 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 2 & 0 & -3 & 1 & 0 & -2 & 0 & 0 & 1 \end{pmatrix}$$

Cusp Equations

Cusp 0:

$$\begin{aligned} \mu_0 &= -Z_1 + Z''_2 \\ \lambda_0 &= 2 Z_1 - 3 Z''_1 + Z_2 - 2 Z''_2 + Z''_3 \end{aligned}$$

3 Edge Classification

Total SnaPy edge equations: 3. Independent edge basis rank: $n - r = 2$. Of these, 0 are **easy** and 2 are **hard**.

3.1 Hard Edges

Hard edges contribute fugacity variables $\eta_0, \eta_1, \dots$ to the refined index.

1. $C_{\text{hard},0} = Z_1 + Z''_1 + Z_2 + 2 Z'_2 + Z_3 + Z''_3$
2. $C_{\text{hard},1} = Z'_1 + Z''_1 + 2 Z''_2 + Z'_3 + Z''_3$

Basis ordering in g_{NZ} (position block rows): meridians (r rows), hard edges, easy edges.

4 Neumann–Zagier Data

4.1 Symplectic Matrix g_{NZ}

Row ordering: meridians $(\mu_0, \dots, \mu_{r-1})$, hard edges, easy edges | longitudes/2 $(\lambda_0/2, \dots)$, Γ vectors.
 Column ordering: $Z_1, \dots, Z_n, Z_1'', \dots, Z_n''$ (block form).

$$g_{NZ} = \begin{pmatrix} -1 & 0 & 0 & 0 & 1 & 0 \\ 1 & -1 & 1 & 1 & -2 & 1 \\ -1 & 0 & -1 & 0 & 2 & 0 \\ 1 & \frac{1}{2} & 0 & -\frac{3}{2} & -1 & \frac{1}{2} \\ -\frac{1}{2} & 0 & -1 & 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & -1 & 0 & 1 & -\frac{3}{2} & 0 \end{pmatrix}$$

4.2 Inverse g_{NZ}^{-1}

$$g_{NZ}^{-1} = \begin{pmatrix} -\frac{3}{2} & 0 & 1 & 0 & -1 & 0 \\ -1 & \frac{1}{2} & -\frac{3}{2} & -1 & 2 & -2 \\ \frac{1}{2} & 0 & 0 & 0 & -1 & 0 \\ -1 & \frac{1}{2} & -\frac{1}{2} & -1 & 1 & -1 \\ -\frac{1}{2} & 0 & 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \end{pmatrix}$$

4.3 Affine Shifts

$$\nu_x = (0, 0, 0)$$

$$\nu_p = (0, 0, 0)$$

5 Local Charges Formula

The refined index is computed by the formula:

$$I^{\text{ref}}(q; \eta_0, \dots, \eta_{k-1}) = \sum_{e_{\text{int}} \in (\frac{1}{2}\mathbb{Z})^{n-r}} \left[\prod_{a=0}^{k-1} \eta_a^{e_{r+a}} \right] \cdot (-q^{1/2})^\varphi \cdot \prod_{j=1}^n I_\Delta((g_{NZ}^{-1}\kappa)_j, (g_{NZ}^{-1}\kappa)_{n+j}) \quad (1)$$

where:

- $\kappa = (m_{\text{ext}}, \underbrace{0, \dots, 0}_{n-r}, e_{\text{ext}}, e_{\text{int}})$ is the $2n$ -vector of external + internal charges.
- The **local charges** (tetrahedron arguments) are $(m_j, e_j) = (g_{NZ}^{-1}\kappa)_j$ and $(g_{NZ}^{-1}\kappa)_{n+j}$.
- The **phase exponent** is $\varphi = m_{\text{full}} \cdot \nu_p - e_{\text{full}} \cdot \nu_x$.
- $I_\Delta(m, e)$ is the tetrahedron index.
- $k = \text{num_hard}$ is the number of hard edges carrying fugacity variables.

6 Non-Closable Cycles & Basis Selection

Cusp 0: non-closable cycles found: $(-1, 0), (1, 0)$.

Selected Basis

- Cusp 0: slope $(1, 0) \Rightarrow m = 1, e = 0$ [default]

7 Dehn Filled Unrefined Indices

The unrefined Dehn-filled index $I_{P/Q}(q)$ is computed for each slope tested in the non-closable cycle search. A slope is *non-closable* when $I_{P/Q}(q)$ is stably zero.

Cusp 0

Stably zero (non-closable): $(-1, 0)$, $(1, 0)$.

P	Q	$I_{P/Q}(q)$
-1	1	$1 - 2q - 3q^2 + 3q^4 + 10q^5$
0	1	$46 - q - 2q^2 - 2q^3 - 4q^4 - 3q^5$
1	1	1

8 Weyl Symmetry & Dehn Filling Compatibility

Convention: $f(\eta) = \eta^{b \cdot m + a \cdot e} \cdot I(m, e)$, $f(\eta) = f(\eta^{-1})$.

$$a = (0, \frac{1}{2}), \quad b = (\frac{1}{2}, 1)$$

- $a \in \mathbb{Z}$: $\times$
- $2b \in \mathbb{Z}$: $\checkmark$

Dehn filling not directly compatible: $a \notin \mathbb{Z}$.

Weyl symmetry check: $\times$ (58/81 entries pass).

9 Refined Index

Evaluated at q -order (half) = 10, giving series up to q^5 .

Output in **Weyl-manifest form**: $f(m, e) = \eta^{b \cdot m + a \cdot e} \cdot I(m, e)$, where the displayed $I(m, e)$ satisfies $I(\eta) = I(\eta^{-1})$.

$$\begin{aligned}
I(-4, -1) &= \eta^{4v_0} \eta^{9v_1} (-q^5 \eta^{8v_0} \eta^{3v_1} - q^5 \eta^{10v_0} \eta^{5v_1}) \\
I(-4, 0) &= \eta^{4v_0} \eta^{8v_1} (q^2 \eta^{8v_0} \eta^{4v_1}) \\
I(-4, 1) &= \eta^{4v_0} \eta^{7v_1} (q^4 \eta^{6v_0} \eta^{3v_1} + q^4 \eta^{8v_0} \eta^{5v_1} + q^5 \eta^{6v_0} \eta^{3v_1} + q^5 \eta^{8v_0} \eta^{5v_1}) \\
I(-4, 2) &= \eta^{4v_0} \eta^{6v_1} (q^5 \eta^{6v_0} \eta^{4v_1}) \\
I(-2, -2) &= \eta^{2v_0} \eta^{6v_1} (-q^4 \eta^{4v_0} \eta^{2v_1} - q^5 \eta^{4v_0} \eta^{2v_1}) \\
I(-2, -1) &= \eta^{2v_0} \eta^{5v_1} (-q^3 \eta^{4v_0} \eta^{v_1} - q^3 \eta^{6v_0} \eta^{3v_1} - q^4 \eta^{4v_0} \eta^{v_1} - q^4 \eta^{6v_0} \eta^{3v_1} - q^5 \eta^{2v_0} \eta^{v_1} - q^5 \eta^{4v_0} \eta^{v_1} - q^5 \eta^{4v_0} \eta^{3v_1} - \\
I(-2, 0) &= \eta^{2v_0} \eta^{4v_1} (q \eta^{4v_0} \eta^{2v_1} - q^4 \eta^{2v_0} - 2q^4 \eta^{4v_0} \eta^{2v_1} - q^4 \eta^{6v_0} \eta^{4v_1} - 2q^5 \eta^{2v_0} - q^5 \eta^{2v_0} \eta^{2v_1} - 4q^5 \eta^{4v_0} \eta^{2v_1} - 2q^5 \eta^{4v_0} \eta^{4v_1}) \\
I(-2, 1) &= \eta^{2v_0} \eta^{3v_1} (q^2 \eta^{2v_0} \eta^{v_1} + q^2 \eta^{4v_0} \eta^{3v_1} + q^3 \eta^{2v_0} \eta^{v_1} + q^3 \eta^{4v_0} \eta^{3v_1} - q^5 \eta^{-v_1} - q^5 \eta^{v_1} - 3q^5 \eta^{2v_0} \eta^{v_1} - q^5 \eta^{2v_0} \eta^{3v_1}) \\
I(-2, 2) &= \eta^{2v_0} \eta^{2v_1} (q^2 \eta^{2v_0} \eta^{2v_1} + q^3 + 2q^3 \eta^{2v_0} \eta^{2v_1} + q^3 \eta^{4v_0} \eta^{2v_1} + q^3 \eta^{4v_0} \eta^{4v_1} - q^4 \eta^{2v_1} + q^4 \eta^{2v_0} \eta^{2v_1} + q^4 \eta^{4v_0} \eta^{2v_1}) \\
I(0, -2) &= \eta^{2v_1} (-q + q^2 \eta^{-2v_0} \eta^{-2v_1} + q^3 \eta^{-2v_0} \eta^{-2v_1} + q^3 \eta^{-2v_1} + 4q^3 + q^3 \eta^{2v_1} + q^3 \eta^{2v_0} + q^3 \eta^{2v_0} \eta^{2v_1} + q^3 \eta^{4v_0} \eta^{2v_1}) \\
I(0, -1) &= \eta^{v_1} (-q \eta^{-v_1} - q \eta^{v_1} - q \eta^{2v_0} \eta^{v_1} + q^2 \eta^{-2v_0} \eta^{-v_1} - q^2 \eta^{2v_0} \eta^{v_1} + 2q^3 \eta^{-2v_0} \eta^{-v_1} + 4q^3 \eta^{-v_1} + 4q^3 \eta^{v_1} + 2q^3 \eta^{2v_0} \eta^{v_1}) \\
I(0, 0) &= 1 - 3q - q \eta^{2v_0} \eta^{2v_1} + q^2 \eta^{-2v_0} \eta^{-2v_1} + q^2 \eta^{-2v_0} - 3q^2 + q^2 \eta^{2v_1} - q^2 \eta^{2v_0} \eta^{2v_1} + 2q^3 \eta^{-2v_0} \eta^{-2v_1} + q^3 \eta^{-2v_1} \\
I(0, 1) &= \eta^{-v_1} (-q \eta^{-v_1} - q \eta^{v_1} - q \eta^{2v_0} \eta^{v_1} + q^2 \eta^{-2v_0} \eta^{-v_1} - q^2 \eta^{2v_0} \eta^{v_1} + 2q^3 \eta^{-2v_0} \eta^{-v_1} + 4q^3 \eta^{-v_1} + 4q^3 \eta^{v_1} + 2q^3 \eta^{2v_0} \eta^{v_1}) \\
I(0, 2) &= \eta^{-2v_1} (-q + q^2 \eta^{-2v_0} \eta^{-2v_1} + q^3 \eta^{-2v_0} \eta^{-2v_1} + q^3 \eta^{-2v_1} + 4q^3 + q^3 \eta^{2v_1} + q^3 \eta^{2v_0} + q^3 \eta^{2v_0} \eta^{2v_1} + q^3 \eta^{4v_0} \eta^{2v_1}) \\
I(+2, -2) &= \eta^{-2v_0} \eta^{-2v_1} (q^2 \eta^{2v_0} \eta^{2v_1} + q^3 + 2q^3 \eta^{2v_0} \eta^{2v_1} + q^3 \eta^{4v_0} \eta^{2v_1} + q^3 \eta^{4v_0} \eta^{4v_1} - q^4 \eta^{2v_1} + q^4 \eta^{2v_0} \eta^{2v_1} + q^4 \eta^{4v_0} \eta^{2v_1})
\end{aligned}$$

$$\begin{aligned}
I(+2, -1) &= \eta^{-2v_0} \eta^{-3v_1} (q^2 \eta^{2v_0} \eta^{v_1} + q^2 \eta^{4v_0} \eta^{3v_1} + q^3 \eta^{2v_0} \eta^{v_1} + q^3 \eta^{4v_0} \eta^{3v_1} - q^5 \eta^{-v_1} - q^5 \eta^{v_1} - 3q^5 \eta^{2v_0} \eta^{v_1} - q^5 \eta^{2v_0} \eta^{3v_1}) \\
I(+2, 0) &= \eta^{-2v_0} \eta^{-4v_1} (q \eta^{4v_0} \eta^{2v_1} - q^4 \eta^{2v_0} - 2q^4 \eta^{4v_0} \eta^{2v_1} - q^4 \eta^{6v_0} \eta^{4v_1} - 2q^5 \eta^{2v_0} - q^5 \eta^{2v_0} \eta^{2v_1} - 4q^5 \eta^{4v_0} \eta^{2v_1} - \\
I(+2, 1) &= \eta^{-2v_0} \eta^{-5v_1} (-q^3 \eta^{4v_0} \eta^{v_1} - q^3 \eta^{6v_0} \eta^{3v_1} - q^4 \eta^{4v_0} \eta^{v_1} - q^4 \eta^{6v_0} \eta^{3v_1} - q^5 \eta^{2v_0} \eta^{v_1} - q^5 \eta^{4v_0} \eta^{v_1} - q^5 \eta^{4v_0} \eta^{3v_1}) \\
I(+2, 2) &= \eta^{-2v_0} \eta^{-6v_1} (-q^4 \eta^{4v_0} \eta^{2v_1} - q^5 \eta^{4v_0} \eta^{2v_1}) \\
I(+4, -2) &= \eta^{-4v_0} \eta^{-6v_1} (q^5 \eta^{6v_0} \eta^{4v_1}) \\
I(+4, -1) &= \eta^{-4v_0} \eta^{-7v_1} (q^4 \eta^{6v_0} \eta^{3v_1} + q^4 \eta^{8v_0} \eta^{5v_1} + q^5 \eta^{6v_0} \eta^{3v_1} + q^5 \eta^{8v_0} \eta^{5v_1}) \\
I(+4, 0) &= \eta^{-4v_0} \eta^{-8v_1} (q^2 \eta^{8v_0} \eta^{4v_1}) \\
I(+4, 1) &= \eta^{-4v_0} \eta^{-9v_1} (-q^5 \eta^{8v_0} \eta^{3v_1} - q^5 \eta^{10v_0} \eta^{5v_1})
\end{aligned}$$